European Polymer Journal 223 (2025) 113632

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Radical polymerization of itaconic acid in dipolar aprotic solvents – The effect of high monomer concentration

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ARTICLE INFO

Keywords:
Radical polymerization
Itaconic acid
Renewable monomers
Kamlet–Taft parameters
Linear solvation energy relationships
Diffusion controlled termination

ABSTRACT

In this study, the dipolar aprotic solvents were used as reaction systems for homopolymerization of itaconic acid (IA). The influence of the dipolar aprotic solvent properties, as represented by the Kamlet-Taft parameters, on the solubility of IA was elucidated using the Linear Solvation Energy Relationships (LSER). Hydrogen bonding in IA solutions was investigated through FT-IR spectroscopy and viscosity measurements. It was found that high IA solubility caused by strong hydrogen bond basicity of the solvents, is a key factor affecting its radical polymerization process, as evidenced by the exponential increase in the molecular weight of the resulting poly(itaconic acid) (PIA) with increasing initial IA concentration. This phenomenon can be explained by diffusion controlled termination of polymerization caused by viscosity of the reaction system. Among the solvents tested, DMSO was identified as the most suitable solvent for IA polymerization due to its high solubility. By using this solvent, average molecular weight of the resulting PIA could be tuned over a wide range from 1 000 to 100 000 g/mol.

1. Introduction

Itaconic acid (IA) is a vinyl monomer of renewable origin, industrially produced by biotechnological methods from biomass feedstock [1]. The presence of a vinyl group and two carboxylic groups in IA makes it a versatile building block suitable for producing both chain-growth polymerization polymers and polyester resins [2–4]. Due to its similar chemical structure, IA is a potential bio-based alternative to petrochemical monomers such as acrylic and methacrylic acids [2–4]. However, the presence of two carboxyl groups presents a steric hindrance to the vinyl group, thus the free-radical polymerization (FRP) of IA proceeds slowly and generally results in low to moderate molecular weight (M_w = 1 000 – 100 000 g/mol) poly(itaconic acid) (PIA) [3–8]. Furthermore, during polymerization, IA undergoes a side reaction of chain transfer to the monomer, which restricts the growth of macroradicals, posing another challenge for the synthesis of high molecular weight PIA [3,5,7,9,10]. Most of the work dealing with IA polymerization has so far focused on processes in aqueous solutions [5–8,11–13]. Despite the outlined difficulties, IA polymerization in aqueous solution using a sufficiently high amount of initiator and an appropriately long reaction time allows for the synthesis of PIA in high yield (> 90 %) [7,11]. Although water is the perfect solvent for the FRP process both for its beneficial effect on reaction kinetics [14] and for being a sustainable solvent [15], the drawback of IA polymerization in water mixtures is the limited solubility of IA, especially at room temperature [16]. Much less work has been devoted to the IA polymerization in organic solvent solutions [6,8,17] in which the solubility of IA is also greatly limited [18].

Significant progress in IA polymerization solving the problems of low solubility of IA was achieved by using so-called Deep Eutectic Solvents (DES) as reaction systems, especially equimolar mixtures of IA and quaternary ammonium chlorides (e.g. choline chloride) [19–25]. The DES systems are formed by the hydrogen bonding between IA serving as a hydrogen bond donor (HBD) and quaternary ammonium chloride acting as a hydrogen bond acceptor (HBA) [26]. The use of DES allows to achieve a higher IA concentration in the reaction system than in the aqueous solution [20] and the IA polymerization proceeds more rapidly than in aqueous solution, which helps to overcome the problem of slow FRP of IA [21]. Furthermore, by using appropriate FRP conditions, i.e. mild temperature and slow initiation rate, the DES systems enable the synthesis of high molecular weight PIA (M_w = 100 000 – 500 000 g/mol) that are one order magnitude higher than the molecular weight of PIA obtained from aqueous solutions [23–25]. In addition to the outlined

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https://doi.org/10.1016/j.eurpolymj.2024.113632
Received 1 October 2024; Received in revised form 4 December 2024; Accepted 7 December 2024
Available online 9 December 2024
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advantages of the IA polymerization in DES, this polymerization also has significant disadvantages: i) the high viscosity of DES makes it difficult to handle such mixtures [22–24]; ii) DES exhibit a supercooling effect and therefore tend to crystallize when handled [22]; iii) PIA undergoes a side reaction in DES i.e. esterification with choline chloride [22,23]; iv) isolation of pure PIA from DES is a laborious multi-step process involving precipitation/dialysis/ion exchange, and the final product may still be contaminated by ammonium salt residues [22,23]. Therefore, it would be beneficial to develop improved method of IA homopolymerization that is less troublesome in use while maintaining the advantages of polymerization in DES.

Inspired by the DES structure containing a strong HBA compound (quaternary ammonium chloride), thanks to which a reaction system with a high IA concentration can be prepared, we decided to try the IA homopolymerization in the presence of dipolar aprotic solvents. Surprisingly, to the best of our knowledge, there are no literature reports on the application of these solvents for the homopolymerization of IA, although there are studies mentioning their successful use for the copolymerization of IA with nonpolar monomers (e.g. styrene, acrylonitrile) [10,27–29].

Therefore, the aim of this work is to investigate the use of dipolar aprotic solvents as a reaction system for homopolymerization of IA. A series of common solvents were considered namely: dimethyl sulfoxide (DMSO), dimethylformamide (DMF), N-methyl-2-pyrrolidone (NMP), trimethyl phosphate (TMeP). The effect of solvent type on IA solubility was correlated with Kamlet-Taft parameters describing the solvent–solute interactions. Hydrogen bonding of IA in dipolar aprotic solvents solutions was investigated with FT-IR spectroscopy and viscosity measurements. The progress of IA polymerization in various mediums (H_2O, DMSO, DMF, NMP, TMeP) and the molecular weight of the resulting PIA were compared. Moreover, the superior IA solubility in DMSO allows for detailed investigation of the effect of high IA concentration (1.0 – 5.0 M) on the polymerization progress and molecular weight of PIA.

2. Experimental

2.1. Materials

IA (≥ 99 %, Sigma-Aldrich), DMSO (≥ 99 %, Sigma-Aldrich), DMF (≥ 99 %, Sigma-Aldrich), NMP (≥ 99 %, Alfa-Aesar), TMeP (≥ 99 %, Sigma-Aldrich) and azobisisobutyronitrile (AIBN, ≥ 98 %, Sigma-Aldrich) were used as received.

2.2. Solubility measurements

IA solubility in dipolar aprotic solvents (DMSO, DMF, NMP, TMeP) was determined at 30 and 65 °C, by the analytical stirred-flask method [18]. The mixture of a given solvent with an excess of IA was stirred overnight at a given temperature to reach equilibrium. Then the saturated IA solution was filtered and its composition was measured using the HPLC method. The measurements were carried out in 9:1 v/v mixture of phosphate buffer (0.1 M Na_2HPO_4) and MeOH as the eluent at 40 °C using Knauer AZURA liquid chromatography system (KNAUER, Germany) with Knauer Azura MWD 2.1L Multi Wavelength UV-Detector. Analyses were conducted using a Phenomenex PolySep-GFC-1000 column according to appropriate linear calibration curve. The accuracy of the method (Tab. S1) was verified against literature data on the solubility of IA (30 °C) and succinic acid (65 °C).

2.3. FT-IR spectroscopy

The FT-IR analyses were performed at room temperature using IRSpirit FT-IR (Shimadzu, Japan) spectrometer equipped with a crystal in horizontal attenuated total reflectance (ATR) cell. Spectra were acquired in a range from 4000 to 400 cm^{-1} with 1 cm^{-1} resolution and average of 16 scans.

2.4. Viscosity measurements

The viscosities were measured using a rotational rheometer RM 200 CP4000 PLUS (Lamy Rheology, France) with cone-plate geometry (diameter of 50 mm, cone angle of 0.5°). The shear rheological experiments were performed in the shear rate range of 800 – 1700 s^{-1}. Measurements were conducted at 25 and 65 °C. The temperature was controlled using the build-in thermostatic system.

2.5. Density measurements

Density measurements were taken using a Kyoto KEM DA-640 (Kyoto Electronics Manufacturing CO., LTD., Japan) vibrating-tube densimeter. The instrument was calibrated at atmospheric pressure using distilled water and dry air.

2.6. Free-Radical polymerization

The sample-specific weights of the reagents (Tab. S2) were placed in a glass vial and dissolved by stirring with a magnetic stir bar at 40 °C. Reagent concentrations were adjusted according to the density of the reaction mixtures (Fig. S1). Then the sample was purged with argon and kept for 48 h at 65 °C while stirred. At appropriate time intervals, samples were taken from the reaction mixture and analyzed by HPLC as described above. The IA conversion was then calculated according to Eq. (1):

$$Conv., \% = \left(1 - \frac{AreaIA_t}{AreaIA_0}\right) \bullet 100\% \quad (1)$$

where: $Area\ IA_0$ is an area of the IA peak on UV chromatogram before polymerization, and $Area\ IA_t$ is an area of the IA peak on UV chromatogram at certain polymerization time.

After 48 h of polymerization, the resulting polymer was purified by dialysis (dialysis tubing Spectra/Por® 7 MWCO 1000) against water and freeze-dried. The polymerization yield was calculated according to Eq. (2):

$$Yield, \% = \frac{m_{PIA}}{m_{IA}} \bullet 100\% \quad (2)$$

where: m_{IA} is a mass of IA in reaction mixture, and m_{PIA} is a mass of resulting PIA after 48 h polymerization. The molecular weight of PIA was determined by Size Exclusion Chromatography (SEC) as previously reported.[24]

2.7. Size exclusion chromatography

SEC was carried out in phosphate buffer (0.1 M Na_2HPO_4) as the eluent at 40 °C using Knauer AZURA liquid chromatography system (KNAUER, Germany) with Knauer Azura RID 2.1L refractive index detector. Analyses were conducted using a PSS SUPREMA XL Linear column calibrated with poly(methacrylic acid) sodium salt standards with narrow molecular weight. Samples for SEC analysis were prepared by dissolving 5 mg of PIA in 1 ml of phosphate buffer at room temperature.

3. Results and discussion

3.1. IA solubility

It is well known that at room temperature the solubility of IA in protic solvents (water, alcohols) and relatively non-polar aprotic solvents (acetone, acetonitrile) is low [16,18]. Table 1 summarizes the literature data on IA solubility in these solvents along with the solubilities of IA in dipolar aprotic solvents (DMSO, DMF, NMP, TMeP)

Table 1
IA solubility expressed as a mass fraction (*wt.%* IA – Eq. S1) and a molar fraction (x_{mol} IA – Eq. S2) in various solvents at 30 °C along with Kamlet-Taft parameters of this solvents.

Solvent	IA solubility			Kamlet-Taft parameters of solvents [33–35]		
	wt.% IA	x_{mol} IA (experimental)	x_{mol}' IA (predicted from the LSER correlation)	α	β	π
H_2O	11.2[16]	0.017[16]	0.030	1.23	0.47	1.14
Methanol	34.1[18]	0.113[18]	0.016	1.00	0.70	0.60
Ethanol	20.2[18]	0.082[18]	0.083	0.86	0.80	0.54
1-Propanol	15.6[18]	0.079[18]	0.132	0.80	0.87	0.52
2-Propanol	17.8[18]	0.091[18]	0.130	0.76	0.88	0.48
Acetone	11.6[18]	0.055[18]	0.065	0.80	0.48	0.71
Acetonitrile	2.1[18]	0.007[18]	0.054	0.15	0.40	0.85
Ethyl acetate	1.8[18]	0.013[18]	−0.002	0	0.45	0.55
TMeP	27.7[1)]	0.292[1)]	0.276	0	0.77	0.72
DMF	45.6[1)]	0.321[1)]	0.283	0	0.69	0.88
NMP	42.1[1)]	0.357[1)]	0.352	0	0.77	0.92
DMSO	49.4[1)]	0.369[1)]	0.376	0	0.76	1.00

[1)] The solubility determined in this work.

measured in this work. It is clearly seen that the IA solubility in dipolar aprotic solvents is significantly higher than in other solvents, including water. A similar trend is observed for other dicarboxylic acids, such as succinic and glutaric acid [30–32]. Differences in the solubility of dicarboxylic acids are associated with the polarity of the respective solvents [31]. More polar solvents are supposed to be more prone to form hydrogen bonds with dicarboxylic acids, so they are able to disrupt the strongly hydrogen-bond-associated crystal structure of dicarboxylic acids [31].

To relate IA solubility (x_{mol} IA) to solvent properties, it is useful to use the Kamlet-Taft three-parameter polarity scale [36–38]. On this scale, the polarity of a solvent is comprehensively characterized by its hydrogen bond acidity (α), hydrogen bond basicity (β) and dipolarity/polarizability (π) [33–35]. Linear solvation energy relationship (LSER, Eq. (3) is known to be applied to correlate a free energy-dependent property (*XYZ*) with solvent Kamlet-Taft parameters by multiple linear regression [33,36].

$$XYZ = XYZ_0 + a \bullet \alpha + b \bullet \beta + c \bullet \pi \tag{3}$$

The parameters (*a*, *b*, *c*) of the LSER equation provide a quantitative measure of the impact of respective Kamlet-Taft parameters on a given *XYZ* property. Solvent effects on various properties such as partition coefficients, reaction rate coefficients and also solubility have been investigated using LSER [32,35]. Thereby, we believe that the use of LSER could help to elucidate the effect of solvent properties on IA solubility (x_{mol} IA).

Therefore, the experimental values of IA solubility (Table 1, x_{mol} IA) were correlated with the solvents Kamlet-Taft parameters (Table 1 – α, β, π) resulting in a fitted LSER (Eq. (4):

$$x'_{mol}IA = -0.5114 - 0.1676 \bullet \alpha + 0.6661 \bullet \beta + 0.3809 \bullet \pi \tag{4}$$

where x_{mol}' IA is IA solubility predicted from the LSER correlation (see SI, Fig. S2, for full regression statistics with upper and lower confidence intervals of the parameters). Briefly, a high R-squared value of 0.9174 indicates a strong fit for the model, which is statistically valid ($F = 0.0001 < 0.05$). Furthermore, all coefficients, including the intercept (XYZ_0), are statistically significant ($P < 0.05$).

The highest value of *b* among the LSER parameters indicates that IA solubility is primarily correlated with the solvent's β parameter, and secondarily with its π parameter. Conversely, since the parameter *a (−0.1676)* has a negative value, there is a negative correlation between the IA solubility and the solvent α parameter. Thus, it seems that dipolar aprotic solvents are better solvents for IA, because they exclusively act as HBA when forming hydrogen bonds with the solute. This is consistent with the literature which claims that carboxylic acids are very prone to form complexes with basic solvents through hydrogen bonds.[39,40] In addition, the obtained LSER correlation (Eq. (4), allows for the prediction of IA solubility (the x_{mol}' IA values in Table 1 and Fig. 1) based on the Kamlet-Taft parameters of the solvent.

3.2. Evaluation of hydrogen bonding in IA solutions by FT-IR spectroscopy and viscosity measurements

The formation of hydrogen bonds by carboxylic acids can be investigated using FT-IR spectroscopy, relying on the shift of the carbonyl group stretching band ($\nu_{(C=O)} = 1750 - 1650\ cm^{-1}$).[41,42] The band of the carbonyl group participating in hydrogen bond formation is shifted to a lower wavenumber, while the band of the non-associated carbonyl group is shifted to a higher wavenumber [41]. IA in its pure crystalline state exhibits a chain-associated structure through hydrogen bonds between carboxyl groups, which correspond to $\nu_{(C=O)\ IA} = 1688\ cm^{-1}$ (Fig. 2c – A) [43]. During the dissolution process, this structure is disrupted and IA-solvent associates are formed. Dipolar aprotic solvents act only as HBA [40], thus their hydrogen bonding interaction with IA carboxyl group is via the hydroxyl group, leaving the carbonyl group of

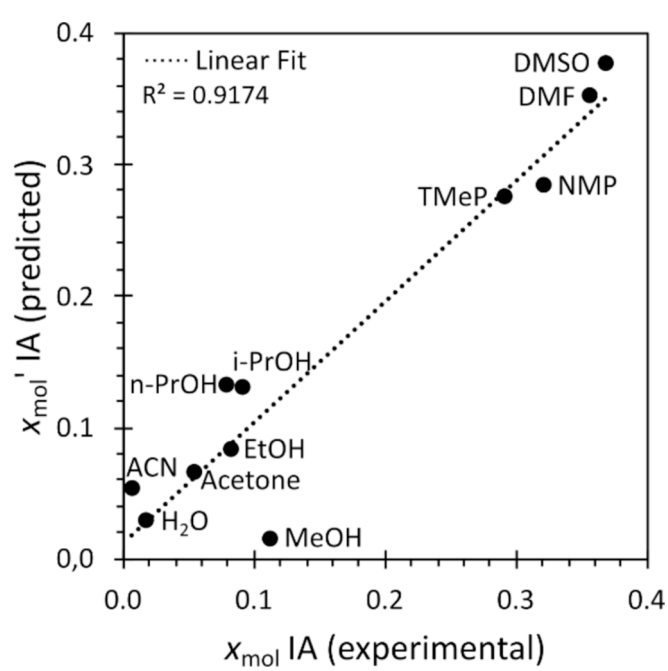

Fig. 1. Cross-correlation of IA solubility predicted by LSER (x_{mol}' IA) and experimentally determined solubility (x_{mol} IA).

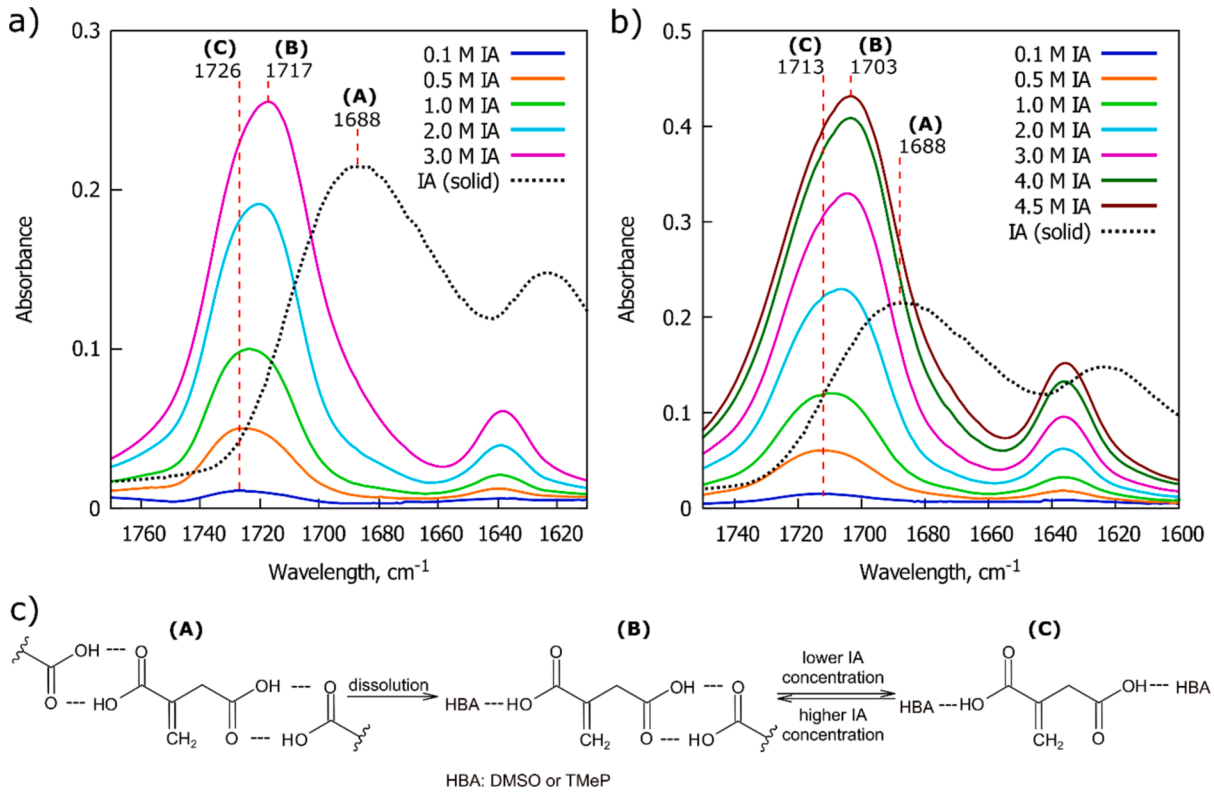

Fig. 2. Regions of FT-IR spectra covering vibrations range of the carbonyl (1750 – 1650 cm^{-1}) and vinyl (1650 – 1600 cm^{-1}) groups of IA in a) TMeP and b) DMSO solutions at different concentrations of IA, compared with solid IA; c) a schematic representation of the effect of IA concentration on hydrogen bonds in the investigated systems: (A) chain-associated structure in solid IA (B) partially associated IA in concentrated solution (C) non-associated IA in diluted solution.

COOH non-associated (Fig. 2c – C). As a result, in dipolar aprotic solvent solution, vibration of the IA carbonyl group is shifted to a lower wavenumber (Fig. 2a – b). IA solutions in DMF and NMP were excluded from the FT-IR studies because these solvents contain a carbonyl group whose vibrations overlap with the IA carbonyl group in the spectra.

Hydrogen bonding of IA in dipolar aprotic solvents is affected not only by the nature of the solvent but also by the IA concentration (Fig. 2a – b). In dilute solutions of IA (0.1 M), the predominant interaction is between carboxyl groups and an excess of TMeP or DMSO giving IA in non-associated state (Fig. 2c – C), as evidenced by the carbonyl group band shift to a higher wavenumber. As the concentration of IA in both TMeP and DMSO solutions increases, the band of the carbonyl groups gradually shifts to a lower wavenumber. This phenomenon can be attributed to an increase in the contribution of intermolecular hydrogen bonds between IA carboxyl groups (Fig. 2c – B) [41]. It appears that as the concentration of IA in solution increases, the hydrogen bonding state increasingly resembles that of the bulk IA exhibiting strong interchain bonding (Fig. 2c – A).

The hydrogen bonding in solutions of carboxylic acids can also be assessed from viscosity measurements [44,45]. Fig. 3 shows the viscosity of IA solutions in dipolar aprotic solvents, which surprisingly increases exponentially with IA concentration. Such a significant increase is not observed in aqueous solutions of dicarboxylic acids, where a linear relationship between viscosity and concentration is usually valid [46,47]. These different viscosity dependences may be attributed to the difference in self-association degree of carboxylic acids, which is substantially greater in less polar organic solvents than in water [39,48]. This hypothesis is consistent with the FT-IR spectroscopy results (Fig. 2), which indicate that IA self-association increases with IA concentration in dipolar aprotic solvents. It seems reasonable to conclude that IA self-association is responsible for the increased viscosity of concentrated IA solutions in dipolar aprotic solvents.

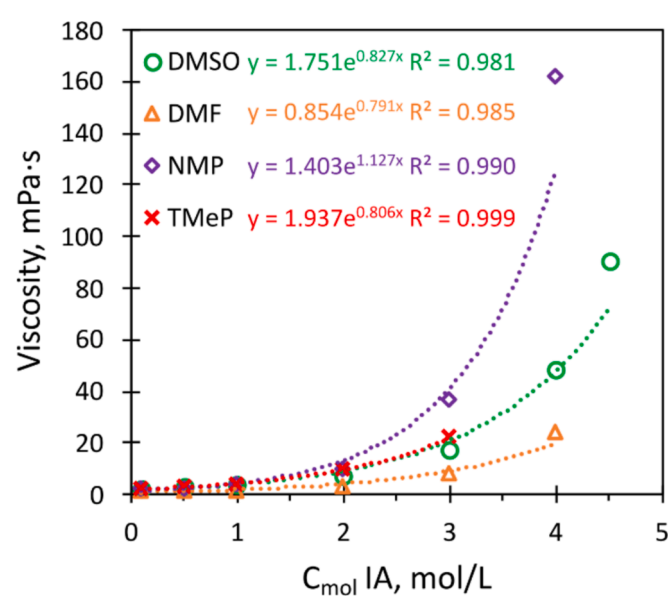

Fig. 3. Viscosities of IA solutions in dipolar aprotic solvents as a function of IA concentration at 25 °C.

3.3. Studies of radical polymerization of IA in various solvents

The radical polymerizations of IA in water, DMSO, DMF, NMP and TMeP were compared. Water is the most common solvent for IA polymerization, so it was selected for comparison with dipolar aprotic solvents. AIBN was used to initiate the polymerization, as it is a versatile radical initiator soluble in organic solvents. Due to its very low solubility

in water, it is uncommon to use AIBN to initiate polymerization in aqueous solution. However, in aqueous IA solution (c_{mol} IA = 3.0 M) the solubility of AIBN is sufficient to achieve the concentration of 0.1 M. The polymerizations were carried out at 65 °C, at which the AIBN decomposes almost completely (~ 98 % – Fig. S3) after 48 h [29]. The same initial IA concentration ($[IA]_0$) of 3.0 M (achievable in all tested solvents, see Tab. S3) was chosen which allowed to compare the polymerization of IA in different solvents.

Fig. 4 shows the conversion curves of IA polymerization in various solvents and Table 2 summarizes these polymerization results. It is clearly visible that IA polymerization in water proceeds faster with higher final monomer conversion than in dipolar aprotic solvents. It is well established that radical polymerization of polar vinyl monomers in water is significantly faster than in organic solvents, and the resulting polymer has a higher molecular weight [14,49]. This is caused by an increased propagation rate coefficient (k_p) due to monomer-water interactions [14,49]. Accordingly, here also the molecular weight of PIA prepared in aqueous solution (M_w = 37 000 g/mol) is much higher than those in other solvents (M_w < 16 000 g/mol).

IA polymerizations in the various dipolar aprotic solvents proceed at similar rates. However, the polymerization in DMSO stands out because the resulting PIA has the second-highest molecular weight after the PIA obtained in aqueous solution (Table 2). It should also be noted that the final monomer conversion varies considerably in different solvents, but never approaches 100 % conversion. This is most likely due to initiator depletion. After 24 h of polymerization, 88 % of the AIBN is decomposed then only ~ 10 % of the AIBN remains for the second day of polymerization (Fig. S3). Consequently, the highest conversion is achieved in aqueous solution, where polymerization rate is the highest before AIBN is depleted. Meanwhile, the very limited solubility of AIBN in IA aqueous solution prevented a comparison of IA polymerization in the respective solvents at a higher concentration of AIBN.

Table 2
Summary of IA polymerization results at 65 °C for 48 h in various solvents at substrates initial concentrations: $[IA]_0$ = 1.0 M, $[AIBN]_0$ = 0.1 M.

Solvent	$[IA]_0$, mol/L	Final IA conversion, %	Resulting PIA			
			Yield, %	M_n, g/mol	M_w, g/mol	Đ
H_2O	3.0	83	77	21 500	37 000	1,7
DMSO	3.0	53	50	8 600	15 800	1,9
DMF	3.0	44	32	6 400	9 800	1,5
NMP	3.0	36	22	5 100	7 600	1,5
TMeP	3.0	38	36	8 800	14 800	1,7

3.4. Effect of IA initial concentration on radical polymerization in DMSO

According to the principles of FRP, both the polymerization rate (Eq. (5) and the molecular weight (Eq. (6) of the resulting PIA should increase with $[IA]_0$.[50] The DMSO solution was chosen to investigate the effect of $[IA]_0$ on IA polymerization. Thanks to exceptionally high solubility of IA in DMSO (Tab. S2), it was possible to increase the $[IA]_0$ up to 5.0 M.

$$R_p \approx \frac{k_p}{\sqrt{k_t}} \bullet \sqrt{f \bullet k_d \bullet [AIBN]} \bullet [IA] \quad (5)$$

$$M_w \propto \frac{k_p}{\sqrt{k_t}} \bullet \frac{1}{\sqrt{f \bullet k_d \bullet [AIBN]}} \bullet [IA] \quad (6)$$

Therefore, the IA polymerization process in DMSO was investigated across the entire range of possible IA initial concentrations ($[IA]_0$ = 1.0 – 5.0 M). Fig. 5a show the conversion curves of IA polymerization in DMSO at various $[IA]_0$ and the results of these polymerizations are summarized in Table 3. It was observed that for $[IA]_0$ > 3.0 M, polymerization proceeds much faster and a significantly higher final monomer conversion is achieved. Variations in final monomer

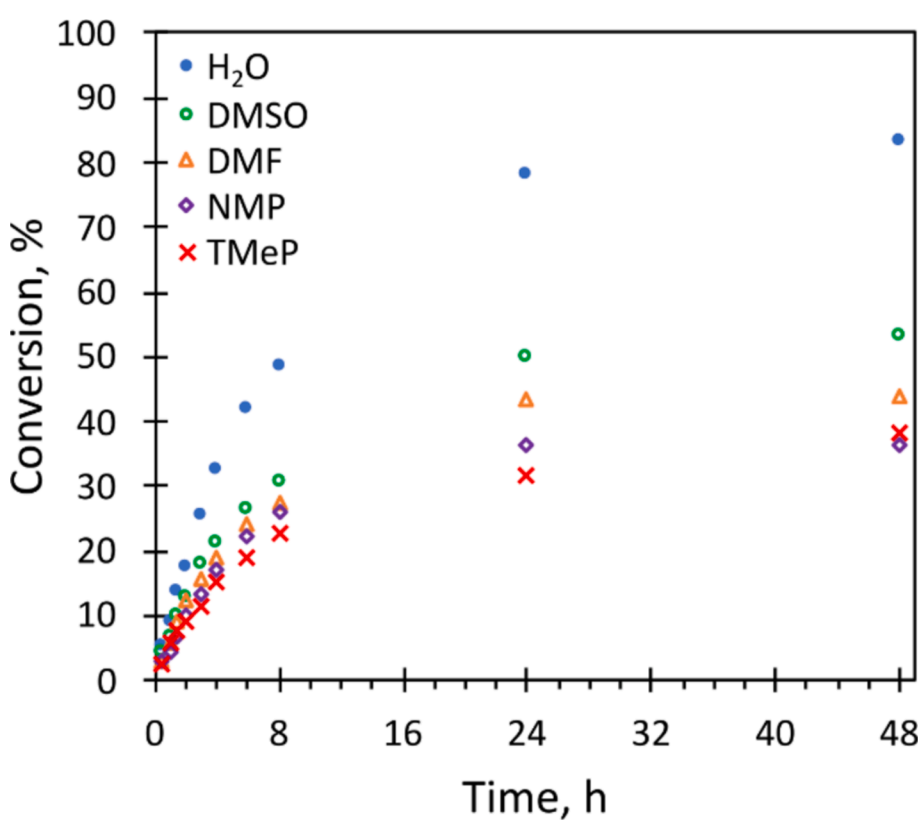

Fig. 4. Conversion curves of IA polymerization at 65 °C in various solvents at substrates initial concentrations: $[IA]_0$ = 3.0 M, $[AIBN]_0$ = 0.1 M.

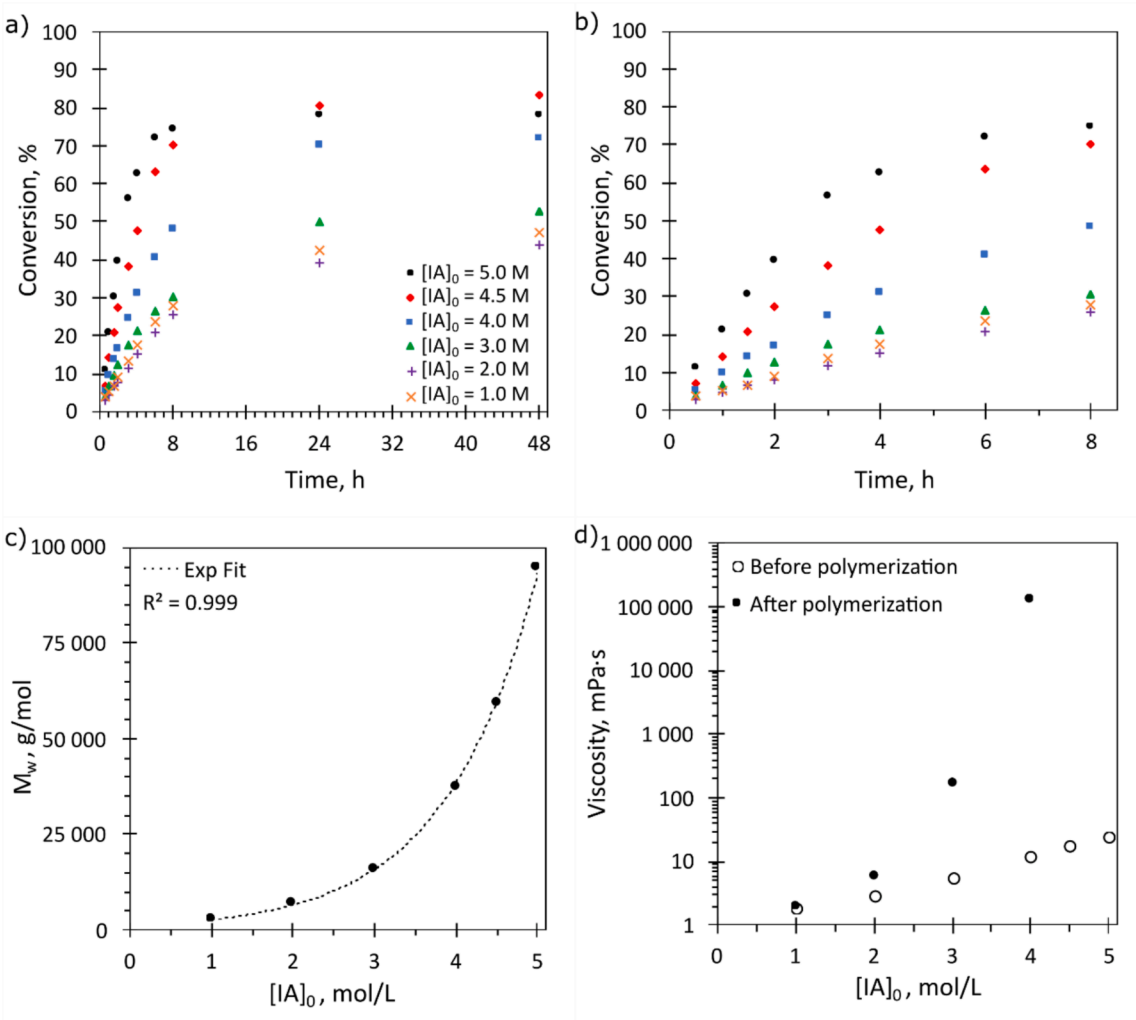

Fig. 5. a) conversion curves of ia polymerization in dmso solutions at 65 °C at substrates initial concentrations: $[IA]_0 = 1.0$ M – 5.0 M, $[AIBN]_0 = 0.1$ M; b) Zoom in on the first 8 h of conversion curves; c) M_w of resulted PIA in function of $[IA]_0$; d) Comparison of viscosity of reaction mixtures at 65 °C before and after polymerization.

Table 3
Summary of IA polymerization results at 65 °C for 48 h in DMSO solution at substrates initial concentrations: $[IA]_0 = 1.0 – 5.0$ M, $[AIBN]_0 = 0.1$ M.

Solvent	$[IA]_0$, mol/L	Final IA Conversion, %	Resulting PIA			
			Yield, %	M_n, g/mol	M_w, g/mol	$Đ$
DMSO	1.0	50	5	1 600	2 600	1.6
	2.0	44	15	3 700	7 000	1.9
	3.0	53	50	8 600	15 800	1.9
	4.0	73	83	20 000	37 500	1.9
	4.5	84	90	33 400	59 100	1.8
	5.0	78	92	53 600	94 900	1.8

conversion have different origins. For the highest IA concentration ($[IA]_0 = 4.5 – 5.0$ M) polymerization almost stops after 8 h, even though a significant amount of AIBN remains in the reaction system (Fig. S3). At such high IA concentrations, these reaction systems vitrify as polymerization progresses, ultimately causing the polymerization to stop. A similar vitrification effect was observed during the polymerization of IA / quaternary ammonium chloride mixtures which are also characterized by high initial concentration of IA.[24] In contrast, the other reaction systems ($[IA]_0 = 1.0 – 4.0$ M) remain liquid after 48 h of polymerization (Fig. 5d). Therefore, the stoppage of their polymerization is related to AIBN depletion as explained earlier for Fig. 4.

Fig. 5c shows that the molecular weight of PIA increases significantly with $[IA]_0$. Therefore, the polymerization in DMSO with the highest concentration of IA ($[IA]_0 = 5.0$ M) produces PIA with a much higher molecular weight ($M_w = 94\,900$ g/mol) than can be obtained by polymerization in aqueous solution (see Table 2). Moreover, the M_w values of the produced PIA exhibit an exponential dependence on $[IA]_0$. This is a surprising result, since according to the kinetics of FRP (Eq. (6), the molecular weight of a polymer should be linearly dependent on $[IA]_0$. As the decomposition rate coefficient (k_d) of AIBN is relatively insensitive to the composition of the reaction mixture [51], a variation in the $k_p/k_t^{0.5}$ ratio with the increasing of $[IA]_0$ is probably responsible for the deviation from the theoretical kinetics.

It is worthwhile to consider the effect of the viscosity on polymerization, as there is an order of magnitude difference in initial viscosity of the reaction mixture in the range of $[IA]_0 = 1.0–5.0$ M (Fig. 5d). An increase in the viscosity of the reaction system leads to diffusion controled termination, which reduces the termination rate coefficient (k_t) [52]. For solution polymerization, it has been proved that as the viscosity increases with increasing monomer concentration, the k_t decreases as a result [53,54]. It is well established that for the diffusion controlled polymerization, the k_t is inversely proportional (Eq. (7) to the viscosity of the reaction system (η) [52,55,56]. Substituting equation (7) into equation (6) gives relationship (Eq. (8), where the molecular weight

of the PIA is linearly dependent on $[IA]_0 \cdot \eta^{1/2}$ [57].

$$k_t = \frac{k_t'}{\eta} \tag{7}$$

$$M_w \propto \frac{k_p}{\sqrt{k_t'}} \bullet \frac{1}{\sqrt{f \bullet k_d \bullet [AIBN]}} \bullet [IA] \cdot \sqrt{\eta} \tag{8}$$

Fig. 6 shows that this relationship is perfectly fitted for range $[IA]_0 = 1.0 - 3.0$ M. This linear correlation indicates that IA polymerization is diffusion controlled even at lower IA concentrations in DMSO. For higher IA concentrations ($[IA]_0 \geq 4.0$ M), the slope of the function gradually increases. This is due to significant change in viscosity during polymerization (Fig. 5d), meaning the initial viscosity does not accurately reflect the change in k_t. For $[IA]_0 \geq 4.0$ M, the decrease in k_t and the corresponding increase in M_w result from both the initial viscosity and the viscosity increase during polymerization.

An analogous effect of $[IA]_0$ on the molecular weight of the resulting PIA can be observed for polymerization in DMF (Fig. S4), although the effect is less pronounced than in DMSO due to the lower solubility of IA in DMF. This suggests that the phenomenon is not exclusive to IA polymerization in DMSO and that it can be utilized to increase the molecular weight of PIA prepared in other reaction systems. To achieve the highest molecular weight of PIA, it is essential to perform IA polymerization in the most concentrated solution possible. This finding is consistent with our previous works in which high molecular weight PIA was obtained in systems with high IA concentrations.[23,24].

4. Conclusions

Dipolar aprotic solvents are particularly effective at dissolving IA, allowing for high IA concentrations that would be unattainable with other solvents. The solubility of IA can be predicted from the Kamlet-Taft parameters of the solvent, which describe its polarity in terms of hydrogen bond donating and accepting ability as well as dipolarity/polarizability. The efficiency of dipolar aprotic solvents in the dissolving of IA is primarily attributed to their strong hydrogen bond accepting ability (hydrogen bond basicity).

Thanks to excellent solubility, dipolar aprotic solvents, particularly DMSO, enable polymerization at a higher initial IA concentration (> 3.0 M) compared to conventional aqueous systems, which results in a higher molecular weight of the PIA. At high IA concentrations within the reaction system, an acceleration of the polymerization rate and an exponential increase in the molecular weight of the resulting PIA are observed. This phenomenon, as evidenced by the viscosity of the reaction system, can be attributed to the diffusion control termination of the polymerization. The results obtained in this work demonstrate that the use of dipolar aprotic solvents for IA polymerization provides outstanding results, and these solvents could serve as an alternative to Deep Eutectic Solvents in the preparation of high molecular weight PIA. However, given that the use of dipolar aprotic solvents in the synthesis processes is environmentally undesirable, a promising direction for further research would be to investigate the IA polymerization in their sustainable substitutes [58,59].

CRediT authorship contribution statement

Kacper Mielczarek: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hynek Beneš:** Writing – review & editing, Writing – original draft, Project administration, Funding acquisition. **Szczepan Bednarz:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

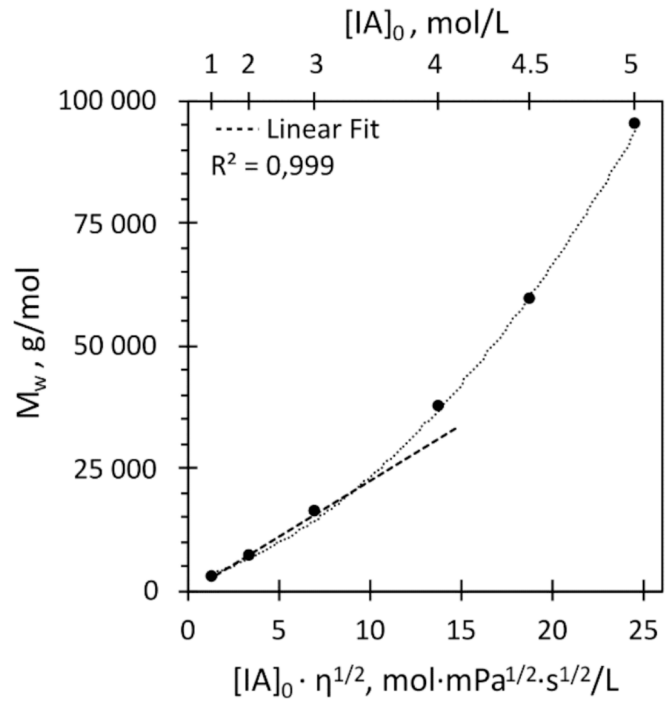

Fig. 6. The correlation between M_w of resulted PIA and $[IA]_0 \cdot \eta^{1/2}$, where $\eta^{1/2}$ is root of viscosity of reaction mixture at 65 °C before polymerization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was financially supported by the Lead Agency bilateral Czech-Polish project provided by the Czech Science Foundation (21–07004K) and National Science Center Poland (CEUS-UNISONO project, grant number 2020/02/Y/ST5/00021).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113632.

Data availability

Data will be made available on request.

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